

10-405

August 24, 2026

1 Syllabus

Languages / Frameworks / Technologies: Python, PySpark, Polars, PyTorch, Amazon EC2

Writing Sessions: Short in-class MCQ-style quizzes that occur after an assignment

Slot A / Slot B: Homework is split into two parts. Slot A does not allow generative AI, and the content is tested in the writing sessions. Slot B is used to correct mistakes made in Slot A, and the max score is kept for each question between the two slots. There is a 10% bonus if the slot A grade is greater than 50%.

- Slot B is mostly optional, and is due 3 days after Slot A grades are released.

2 Scaling Laws & Big Data

Compute Scaling = Data Scaling: Scaling compute (e.g., model size) by a factor of k is approximately equivalent to scaling the data by the same factor of k , in terms of reduction in loss.

Naive Bayes classifier: Consider performing a binary classification problem (e.g., affect vs effect), we predict

$$\hat{y} = \operatorname{argmax}_y \mathbf{P}\{y\} \prod_{i=1}^d \mathbf{P}\{x_i | y\} = \operatorname{argmax}_y \log \mathbf{P}\{y\} + \sum_{i=1}^d \log \mathbf{P}\{x_i | y\}$$

where

$$\mathbf{P}\{x | y\} = \frac{n(\text{word } x \text{ appears in doc with label } y) + 1/V}{n(\text{words in corpus with label } y) + 1}.$$